TikTok Viral Detector

NLP Model Architecture · Training Data · Feature Weights
Tatiana DeMaria — Full Catalog Viral Analysis

Model 1: Line Classifier

Model 2: Similarity

Model 3: Song Virality

How the System Works

INPUT

User pastes any song lyrics (or Tatiana's CSV is loaded directly)

PARSE

Section headers detected & stripped (Chorus, Verse, Bridge, etc.)

MODEL 1

Every lyric line scored 0–100%
Gradient Boosting on 11 features

MODEL 2

Full song scored 0–100%
13 aggregated song-level features

OUTPUT

Viral line + TikTok clip
Song score + ranked table

Training Data

Summer Dataset

4,785 songs · 5 tiers (0–4)
Tier 4 = Super Hits

Genius Lyrics API

Lyrics fetched for 231 songs
Lyric text = model input

TikTok Trending

live trending songs
Labelled viral = 1

Model 1 — Viral Line Classifier

Hyperparameters & Results

Algorithm	Gradient Boosting Classifier
n_estimators	300 trees
max_depth	4
learning_rate	0.08
subsample	0.85
Training data	12,282 lyric lines
Classes	Viral (1) / Non-Viral (0)
CV F1 Score	0.845 ± 0.008
Accuracy	87.3%

Weak Supervision Label Rule:

Line is VIRAL (y=1) if it is in [Chorus/Hook/Bridge] section OR repeats ≥ 3× in the song

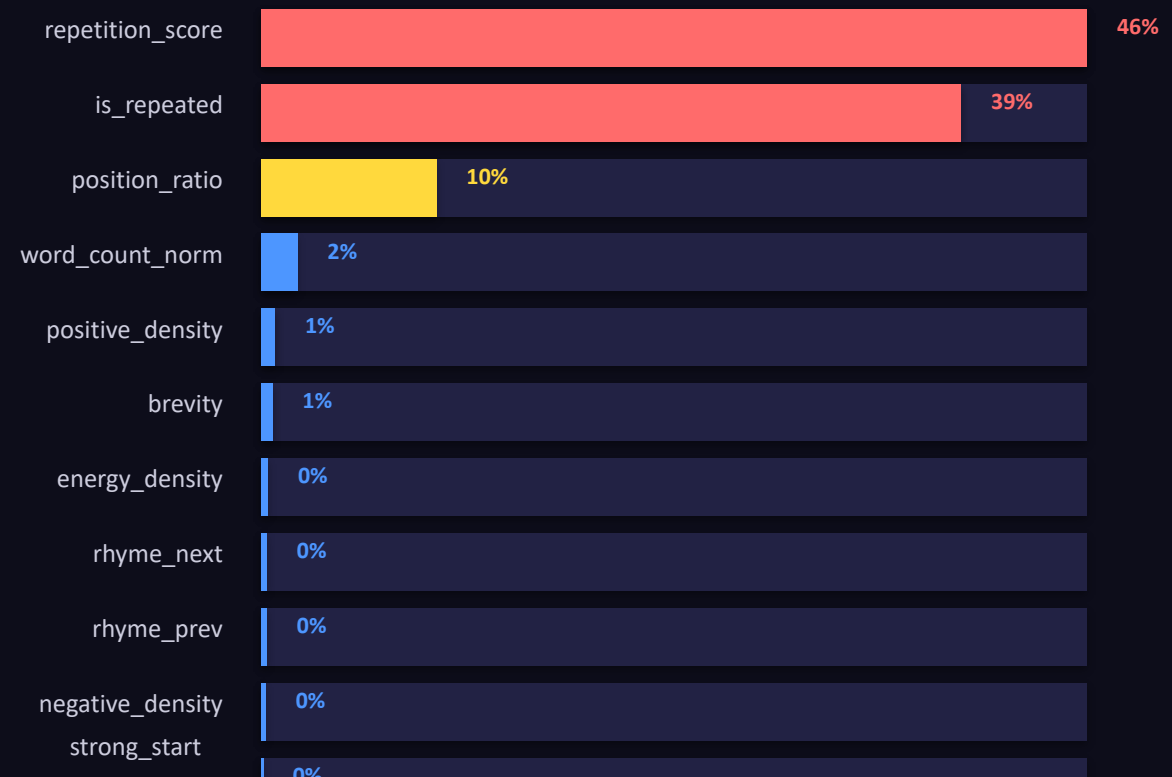
Training Set Breakdown

12,282 lyric lines total

49% Viral lines (6,000)

51% Non-viral lines (6,282)

Feature Importance Weights (Model 1)

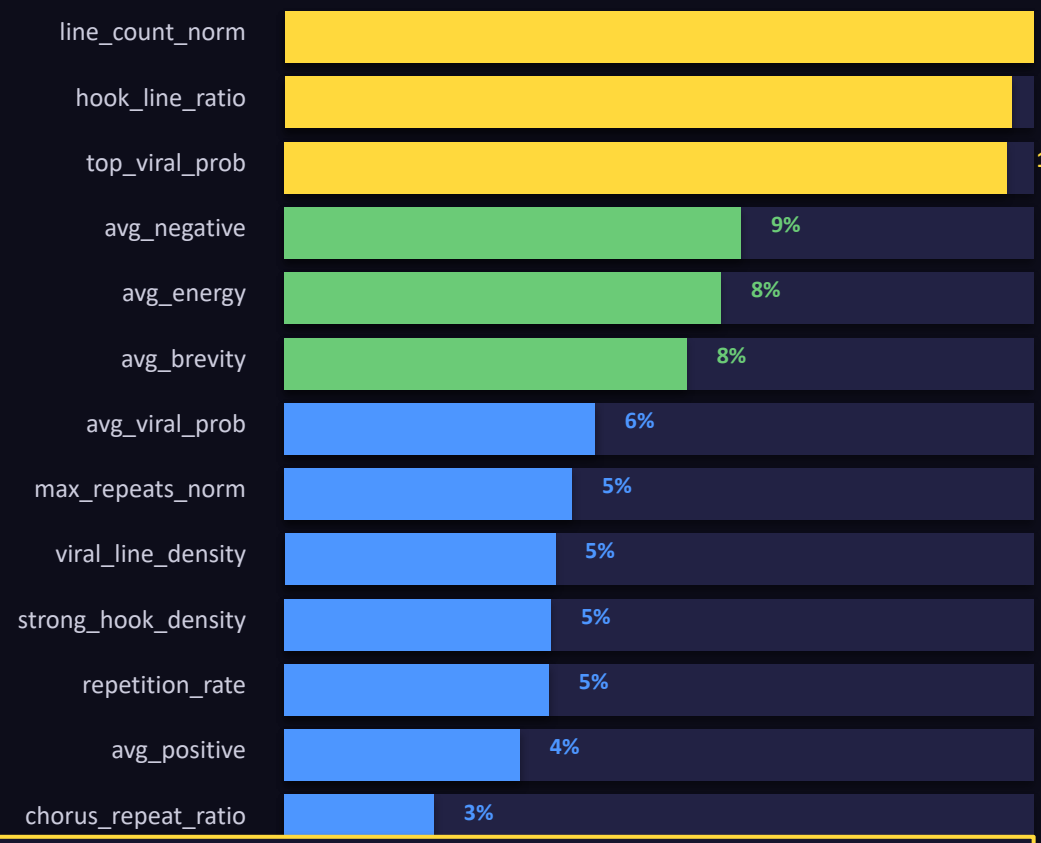


Model 2 — Song-Level TikTok Virality Predictor

Hyperparameters & Label Design

Algorithm	Gradient Boosting Classifier
n_estimators	200 trees
max_depth	3
learning_rate	0.08
subsample	0.80
Training songs	187 songs (tiers 0,1,3,4,-1)
Label 1 (viral)	Tier 3/4 + TikTok trending (tier -1)
Label 0 (not)	Tier 0/1 (non-hits / minor hits)
Tier 2	Skipped (ambiguous mid-tier)
CV F1 Score	0.678 ± 0.057

Feature Importance Weights (Model 3)



Dependency Chain:

Model 1 must be trained first. Model 3 calls Model 1 on every training song to compute avg_viral_prob, top_viral_prob, and density features — these song-level aggregates are then fed into Model 3.

Tatiana DeMaria — Catalog Overview

6	2	4	58%	12
🔥 High Potential (≥65%)	⚡ Moderate (45–64%)	📉 Low (<45%)	Avg Viral Score	Songs Analysed

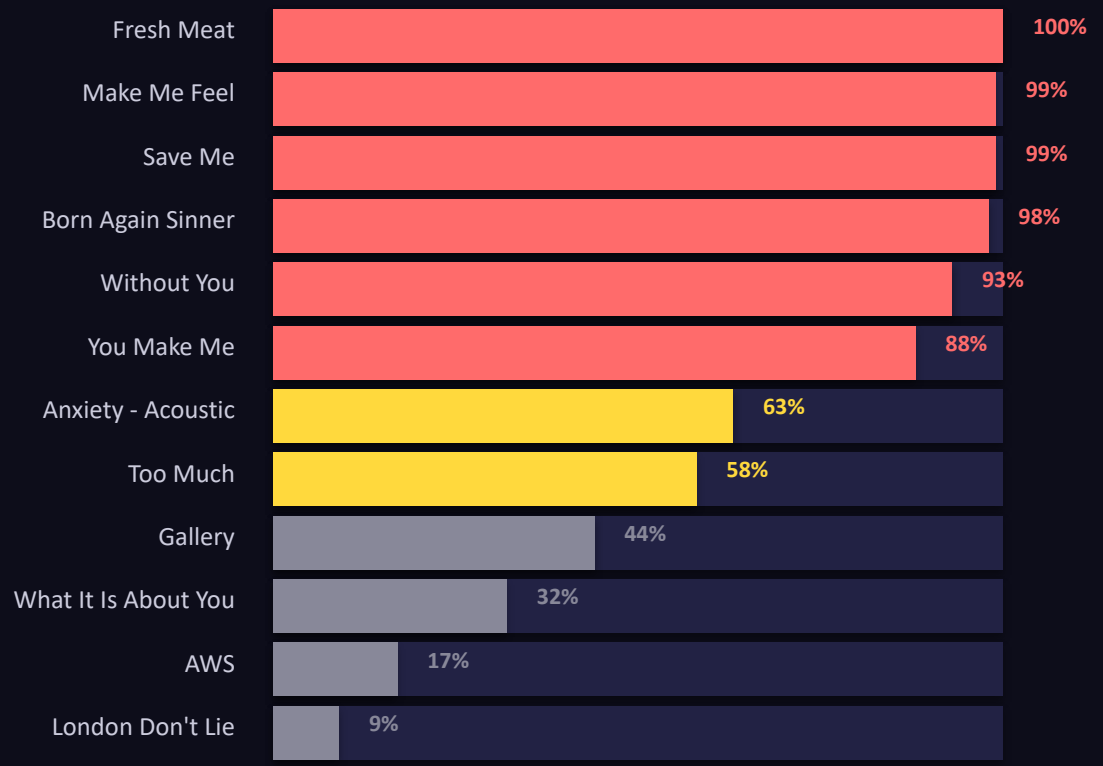
Top Pick

Fresh Meat

TikTok Viral Score: 100% · 🔥 High TikTok Potential

"Rotting here down in the belly of the beast"

Score Distribution



Tatiana DeMaria — Full Ranked Table

#	Song	Score	Rating	Viral Line	Section	Line %
1	**Fresh Meat**	100%	🔥	*"Rotting here down in the belly of the beast"*	[Verse]	97%
2	**Make Me Feel**	99%	🔥	*"Caught here"*	[Chorus]	82%
3	**Save Me**	99%	🔥	*"You'll be waiting for someone to save you"*	[Chorus]	88%
4	**Born Again Sinner**	98%	🔥	*"I'm a born again sinner"*	[Chorus]	94%
5	**Without You**	93%	🔥	*"But see without you it means nothing"*	[Chorus]	90%
6	**You Make Me**	88%	🔥	*"You, You Make Me"*	[Chorus]	96%
7	**Anxiety - Acoustic**	63%	⚡	*"C'mon and try and shake it"*	[Chorus]	75%
8	**Too Much**	58%	⚡	*"Too much it ties me up"*	[Chorus]	89%
9	**Gallery**	44%	📉	*"when you're afraid and all alone?"*	[Tag]	71%
10	**What It Is About You**	32%	📉	*"With your guilt free heart"*	[Chorus]	97%
11	**AWS**	17%	📉	*"I finally found my way into my mind"*	[Chorus]	78%
12	**London Don't Lie**	9%	📉	*"No don't lie no no"*	[Chorus]	68%

Score = Model 3 song-level viral probability · Line % = Model 1 viral probability for the top clip line · Section = where the viral line lives

Top 3 Songs — Viral Line Spotlight

#1 Fresh Meat

🔥 100%

VIRAL LINE TO FILM

"Rotting here down in the belly of the beast"

Line Viral Prob	97%
Best Section	[Verse]
Song Score	100%

Why it will go viral:

High energy language + strong repetition rate. The line "belly of the beast" is visceral and clip-worthy.

#2 Make Me Feel

🔥 99%

VIRAL LINE TO FILM

"Caught here"

Line Viral Prob	82%
Best Section	[Chorus]
Song Score	99%

Why it will go viral:

Ultra-short viral line (2 words). Appears in Chorus. POV / lip sync format is obvious.

#3 Save Me

🔥 99%

VIRAL LINE TO FILM

"You'll be waiting for someone to save you"

Line Viral Prob	88%
Best Section	[Chorus]
Song Score	99%

Why it will go viral:

Emotionally resonant hook. 88% viral probability — strong repetition drives duet culture.

Key Findings & Recommendations



6 / 12 songs have High TikTok Potential

Fresh Meat, Make Me Feel, Save Me, Born Again Sinner, Without You, and You Make Me all score ≥88%.



Repetition is the strongest signal

Model 1 gives 46% weight to repetition_score and 39% to is_repeated. Songs where a hook line repeats ≥3× are consistently flagged viral.



Chorus lines dominate across all songs

10 of 12 top viral lines live in the Chorus section. Prioritise chorus polish for maximum TikTok impact.



Short lines outperform long lines

"Caught here" (2 words, 99% song score) shows brevity wins. Aim for 4–9 words in the hook line.



London Don't Lie scores lowest (9%)

Low repetition rate, complex structure, limited hook density. Consider a stripped-back acoustic version for TikTok.